Homework 7

● Graded

Student

Colin Skinner

Total Points

100 / 100 pts

Question 1

Problem 1

25 / 25 pts

✔ **– 0 pts** Correct

– 1 pt Minor Error

– 3 pts Incorrect Inertial Integration

– 4 pts Did not find the rotated tensor

– 2 pts Wrong Transformation

Question 2

Problem 2

25 / 25 pts

✔ **– 0 pts** Correct

– 1 pt minor error

– 2 pts Incomplete algebra

Question 3

Problem 3

25 / 25 pts

✔ **– 0 pts** Correct

– 3 pts Minor Error

– 5 pts Correct conservations but larger error/incomplete

– 5 pts Didn't solve for omega

Question 4

Problem 4

25 / 25 pts

✔ **– 0 pts** Correct

– 5 pts Does not state conservations

– 3 pts Missing some justification on conservations

– 20 pts No attempt

– 10 pts Does not attempt a solution

Question assigned to the following page: 1

ellipsoid of density ρ about its center of mass. The ellipsoid has semi-major axes a, b, and c in the x, y, and z directions, respectively. Express your answer in terms of the ellipsoid mass, m. Then find the inertia tensor resolved in a frame rotated by θ about the principal y-axis. What happens when $a=c$? What does this imply about the rotated frame? (Hint! Try using symmetry arguments and change of variables techniques to simplify the integration)

$$I_{\alpha\beta} = \int_V \rho \left[\delta_{\alpha\beta} r^2 - r_\alpha r_\beta \right] \qquad \delta_{\alpha\beta} = \begin{cases} 0 & \alpha \neq \beta \\ 1 & \alpha = \beta \end{cases}$$

for ellipsoid: $x = ar\sin\phi\cos\theta \quad y = br\sin\phi\sin\theta \quad z = cr\cos\phi$

$$I_{xx} = \int_V \rho\left[r^2 - x^2\right] dV = \rho \int_V \left(x^2+y^2+z^2 - x^2\right) dV = \rho \int_V \left(y^2+z^2\right) dx\, dy\, dz$$

$$= \rho \int_V \left(y^2+z^2\right) \det|J|\, dr\, d\phi\, d\theta$$

$$= \rho\, abc \int_0^1 \int_0^{2\pi} \int_0^{\pi} \left(b^2 r^2 \sin^2\phi \sin^2\theta + c^2 r^2 \cos^2\phi\right) r^2 \sin\phi \, d\phi\, d\theta\, dr$$

$$= \rho\, abc \int_0^1 r^4 dr \int_0^{2\pi} \int_0^{\pi} b^2 \sin^3\phi \sin^2\theta + c^2 \cos^2\phi \sin\phi \; d\phi\, d\theta$$

$\frac{1}{5}$ ①

① $$= \int_0^{\pi} b^2 \sin^3\phi \sin^2\theta + c^2 \cos^2\phi \sin\phi \; d\phi = \int_0^{\pi} b^2 \left(1-\cos^2\phi\right) \sin\phi \sin^2\theta + c^2 \cos^2\phi \sin\phi \; d\phi$$

$$= \int_0^{\pi} b^2 \sin\phi \sin^2\theta - b^2 \cos^2\phi \sin\phi \sin^2\theta + c^2 \cos^2\phi \sin\phi \; d\phi$$

$u = \cos\phi \quad du = d$

$$= b^2 \sin^2\theta \int_0^{\pi} \sin\phi \, d\phi + \left(-b^2\sin^2\theta + c^2\right) \int_0^{\pi} \cos^2\phi \sin\phi \, d\phi$$

$$= -b^2 \sin^2\theta \cos\phi \Big|_0^{\pi} - \left(-b^2\sin^2\theta + c^2\right) \int_1^{-1} u^2 du = 2b^2\sin^2\theta + \left(b^2\sin^2\theta - c^2\right)\left(\tfrac{1}{3}u^3\right)\Big|_1^{-1}$$

$$= 2b^2\sin^2\theta - \left(-b^2\sin^2\theta + c^2\right)\left(\tfrac{1}{3}u^3\right)\Big|_1^{-1} = 2b^2\sin^2\theta + \tfrac{2}{3}\left(-b^2\sin^2\theta + c^2\right)$$

$= \left(\frac{4}{3}\right) b \sin\theta + \frac{2}{3}c = \frac{4}{3}b^2\sin^2\theta + \frac{2}{3}c^2$

$$I_{xx} = \rho abc \frac{1}{5} \int_0^{2\pi} \frac{4}{3}b^2\sin^2\theta + \frac{2}{3}c^2 \, d\theta$$

②

$$② = \int_0^{2\pi} \frac{4}{3}b^2\sin^2\theta + \frac{2}{3}c^2 d\theta = \int_0^{2\pi} \frac{4}{3}b^2\frac{1}{2}\left(1-\cos(2\theta)\right) + \frac{2}{3}c^2 d\theta$$

$$= \frac{2}{3}b^2\left(\theta - \frac{1}{2}\sin(2\theta)\right) + \frac{2}{3}c^2\theta \Big|_0^{2\pi} = \frac{4\pi}{3}b^2 + \frac{4\pi}{3}c^2$$

0 from $0 - 4\pi$

$$I_{xx} = \rho\frac{1}{5}abc\left(\frac{4\pi}{3}b^2 + \frac{4\pi}{3}c^2\right) \qquad I_{xx} = \frac{4}{15}\pi\rho abc\left(b^2+c^2\right)$$

$$I_{yy} = \int_V \rho\left[r^2 - y^2\right]dV = \rho\int_V \left(x^2+y^2+z^2-y^2\right)dV = \rho\int_V \left(x^2+z^2\right)dV$$

basically same as before

$$= \rho abc \int_V \left(x^2+z^2\right)dV = \rho abc \int_0^1\int_0^{2\pi}\int_0^{\pi} \left(a^2v^2\sin^2\phi\cos^2\theta + c^2v^2\cos^2\phi\right)v^2\sin\phi \, d\phi \, d\theta \, dv$$

$$= \rho abc \int_0^1 r^4 dr \int_0^{2\pi}\int_0^{\pi} a^2\sin^3\phi\cos^2\theta + c^2\cos^2\phi\sin\phi \, d\phi d\theta$$

③

θ is constant across ϕ so this is the same result as ① and with a^2 not b^2

$$I_{yy} = \rho abc \frac{1}{5}\int_0^{2\pi} \frac{4}{3}a^2\cos^2\theta + \frac{2}{3}c^2 \, d\theta$$

④

(4) $= \int_0^{2\pi} \frac{4}{3} a^2 \cos^2\theta + \frac{2}{3} c^2 \, d\theta = \int_0^{2\pi} \frac{4}{3} a^2 \frac{1}{2}\left(1+\cos(2\theta)\right) + \frac{2}{3} c^2 \, d\theta$

$= \frac{2}{3} a^2 \left(\theta + \frac{1}{2}\sin(2\theta)\right) + \frac{2}{3} c^2 \theta \Big|_0^{2\pi} = \frac{4\pi}{3} a^2 + \frac{4\pi}{3} c^2$ (0 from $\theta = 4\pi$)

$I_{xx} = \rho abc \frac{1}{5}\left(\frac{4\pi}{3} a^2 + \frac{4\pi}{3} c^2\right)$ $\qquad$ $I_{yy} = \frac{4}{15}\pi \rho abc \left(a^2 + c^2\right)$

$I_{zz} = \int_V \rho\left[r^2 - z^2\right] dV = \rho \int_V \left(x^2+y^2+z^2-z^2\right) dV = \rho \int_V \left(x^2+y^2\right) dV$

$= \rho abc \int_V \left(x^2+y^2\right) dV = \rho abc \int_0^1 \int_0^{2\pi} \int_0^{\pi} \left(a^2 r^2 \sin^2\phi \cos^2\theta + b^2 r^2 \sin^2\phi \sin^2\theta\right) r^2 \sin\phi \, d\phi \, d\theta \, dr$

$= \rho abc \int_0^1 r^4 \, dr \int_0^{2\pi} \left(a^2\cos^2\theta + b^2\sin^2\theta\right) d\theta \int_0^{\pi} \sin^3\phi \, d\phi$

(1) $= \int_0^{2\pi} \frac{1}{2}a^2\left(1+\cos 2\theta\right) + \frac{1}{2}b^2\left(1-\cos 2\theta\right) d\theta = \frac{1}{2}a^2\left(\theta + \frac{1}{2}\sin 2\theta\right) + \frac{1}{2}b^2\left(\theta - \frac{1}{2}\sin 2\theta\right) \Big|_0^{2\pi}$ (0 and 0)

$= \frac{1}{2}a^2(2\pi) + \frac{1}{2}b^2(2\pi) = \pi\left(a^2+b^2\right)$

(2) $= \int_0^{\pi} \sin^3\phi \, d\phi = \int_0^{\pi} \left(1-\cos^2\phi\right)\sin\phi \, d\phi = \int_0^{\pi} \sin\phi - \cos^2\phi \sin\phi \, d\phi$

$= \int_0^{\pi} \sin\phi \, d\phi - \int_0^{\pi} \cos^2\phi \sin\phi \, d\phi = -\cos\phi \Big|_0^{\pi} + \int_1^{-1} u^2 \, du$

Question assigned to the following page: 1

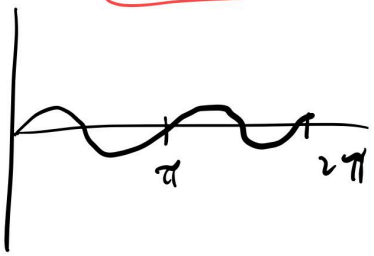
$= 2 + \frac{1}{3}(-1-1) = 2 - \frac{2}{3} = \frac{4}{3}$

$I_{zz} = \rho\left(\frac{1}{5}\right)abc\,\pi\left(a^2+b^2\right)\frac{4}{3}$ $\qquad$ $I_{zz} = \frac{4}{15}\rho abc\,\pi\left(a^2+b^2\right)$

off diagonal

$$I_{xy} = \int_V \rho[-xy]\,dV = \rho abc \int_0^1\int_0^{2\pi}\int_0^{\pi} (ar\sin\phi\cos\theta)(br\sin\phi\sin\theta)\,r^2\sin\phi\,d\phi\,d\theta\,dr$$

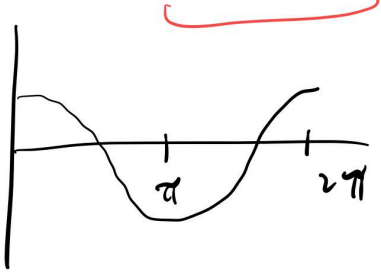
$$= \rho ab\alpha \int_0^1 r^4\,dr \int_0^{2\pi} \sin\theta\cos\theta\,d\theta \int_0^{\pi} \sin^3\phi\,d\phi \qquad I_{xy} = I_{yx} = 0$$



Integrating from $0 \to 2\pi$ gives area of 0

$$I_{xz} = \int_V \rho[-xz]\,dV = \rho\int_0^1\int_0^{2\pi}\int_0^{\pi} (ar\sin\phi\cos\theta)(cr\cos\phi)\,r^2\sin\phi\,d\phi\,d\theta\,dr$$

$$= \rho ac \int_0^1 r^4\,dr \int_0^{2\pi} \cos\theta\,d\theta \int_0^{\pi} \sin^2\phi\cos\phi\,d\phi \qquad I_{xz} = I_{zx} = 0$$



Integrating from $0 \to 2\pi$ gives area of 0

Same with yz because $y \sim \cos\theta$ with no other θ terms

so $I_{yz} = I_{zy} = 0$

Question assigned to the following page: 1

Volume $(x,y,z) \to (x',y',z')$

$$J = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix}$$

$\det J = abc$

$$V = \int dV = \int_0^1 \int_0^{2\pi} \int_0^{\pi} abc\, r^2 \sin\phi \, d\phi \, d\theta \, dr$$

$$= abc \int_0^1 r^2 dr \int_0^{2\pi} d\theta \int_0^{\pi} \sin\phi \, d\phi = abc \left[\tfrac{1}{3} r^3 \Big|_0^1\right](2\pi)\left(-\cos\phi \Big|_0^{\pi}\right)$$

$$= abc\left(\tfrac{1}{3}\right)(2\pi)(1+1) = \frac{4}{3}\pi abc \quad \to \quad \rho = \frac{m}{V} = \frac{3m}{4\pi abc}$$

$$I_{xx} = \frac{4}{15}\pi \rho (b^2+c^2) = \frac{4}{5}\pi \left(\frac{3m}{4\pi abc}\right)(b^2+c^2)$$

$$\{I\} = \frac{3}{5} m \begin{bmatrix} b^2+c^2 & 0 & 0 \\ 0 & a^2+c^2 & 0 \\ 0 & 0 & a^2+b^2 \end{bmatrix}$$

$$R_y(\theta) = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \qquad \{I'\} = R\{I\}R^T$$

$$= \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \frac{3}{5} m \begin{bmatrix} b^2+c^2 & 0 & 0 \\ 0 & a^2+c^2 & 0 \\ 0 & 0 & a^2+b^2 \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

Question assigned to the following page: 1

$$[I'] = \frac{3M}{5}\begin{bmatrix} (b^2+c^2)\cos\theta & 0 & -(a^2+b^2)\sin\theta \\ 0 & a^2+c^2 & 0 \\ (b^2+c^2)\sin\theta & 0 & (a^2+b^2)\cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$$

$$[I'] = \frac{3M}{5}\begin{bmatrix} (b^2+c^2)\cos^2\theta + (a^2+b^2)\sin\theta & 0 & (c^2-a^2)\cos\theta\sin\theta \\ 0 & a^2+c^2 & 0 \\ (c^2-a^2\cos\theta\sin\theta & 0 & (b^2+c^2)\sin^2\theta - (a^2+b^2)\cos^2\theta \end{bmatrix}$$

when $a=c$, matrix becomes diagonal, which means that any rotation around y will not change the inertia tensor. Intuitively, this makes sense because it turns the ellipsoid into a "bulge-y" plate centered with symmetry around the y axis.

Question assigned to the following page: 2

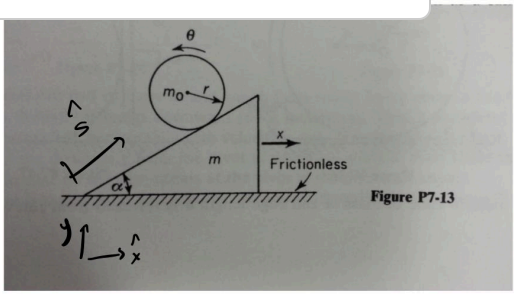


Figure P7-13

A solid homogeneous sphere of mass m_0 and radius r rolls without slipping on a triangular block of mass m which can slide on a frictionless floor. Assuming that the system is initially motionless, solve for velocity of the block as a function of time.

$q: s, x$ ← x of block

constraints: $\dot{s} = r\dot{\theta}$ (1)

$V_{sphere} = m_0 g (y - y_0)$ (y_0 potential energy)

$= m_0 g s \cdot \sin\alpha$

$T_{block} = \frac{1}{2} m \dot{x}^2$ $\quad T_{sp} = \frac{1}{2} I \dot{\theta}^2 = \frac{1}{2} \cdot \frac{2}{5} m_0 r^2 \left(\frac{\dot{s}}{r}\right)^2 = \frac{1}{5} m_0 \dot{s}^2$

$v_{sp} = (\dot{x} + \dot{s}\cos\alpha)\hat{x} + (\dot{s}\sin\alpha)\hat{y}$

$T_{sp} = \frac{1}{2} m_0 \left[(\dot{x} + \dot{s}\cdot\cos\alpha)^2 + (\dot{s}\cdot\sin\alpha)^2\right]$

$= \frac{1}{2} m_0 \left[\dot{x}^2 + 2\dot{x}\dot{s}\cos\alpha + \dot{s}^2\cos^2\alpha + \dot{s}^2\sin^2\alpha\right]$

$= \frac{1}{2} m_0 \left[\dot{x}^2 + 2\dot{x}\dot{s}\cos\alpha + \dot{s}^2\right]$

$\mathcal{L} = T - V$

$\mathcal{L} = \frac{1}{2} m \dot{x}^2 + \frac{1}{2} m_0 \dot{x}^2 + m_0 \dot{x}\dot{s}\cos\alpha + \frac{1}{2} m_0 \dot{s}^2 + \frac{1}{5} m_0 \dot{s}^2 + m_0 g s \cdot \sin\alpha = 0$

$= \frac{1}{2}(m + m_0)\dot{x}^2 + m_0 \dot{x}\dot{s}\cos\alpha + \frac{7}{10} m_0 \dot{s}^2 - m_0 g s \cdot \sin\alpha = 0$

$\frac{\partial \mathcal{L}}{\partial \dot{x}} = (m + m_0)\dot{x} + m_0 \dot{s}\cos\alpha$ $\quad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{x}} = (m + m_0)\ddot{x} + m_0 \ddot{s}\cos\alpha$

$\frac{\partial \mathcal{L}}{\partial x} = 0$ $\quad (m + m_0)\ddot{x} + m_0 \ddot{s}\cos\alpha = 0$

$\frac{\partial \mathcal{L}}{\partial \dot{s}} = m_0 \dot{x}\cos\alpha + \frac{7}{5} m_0 \dot{s}$ $\quad \frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{s}} = m_0 \ddot{x}\cos\alpha + \frac{7}{5} m_0 \ddot{s} = 0$

$\frac{\partial \mathcal{L}}{\partial s} = -m_0 g \sin\alpha$ $\quad m_0 \ddot{x}\cos\alpha + \frac{7}{5} m_0 \ddot{s} + m_0 g \sin\alpha = 0$

Question assigned to the following page: 2

$$\ddot{s} = -\frac{(M+M_0)}{M_0\cos\alpha}\ddot{x} \rightarrow M_0\ddot{x}\cos\alpha - \frac{2}{5}M_0\frac{(M+M_0)}{M_0\cos\alpha}\ddot{x} + M_0 g\sin\alpha = 0$$

$$\ddot{x}\left[\frac{2}{5}\frac{M+M_0}{\cos\alpha} - M_0\cos\alpha\right] = M_0 g\sin\alpha$$

$$\int_0^{\dot{x}} \ddot{x}\left[\frac{2}{5}(M+M_0) - M_0\cos^2\alpha\right]dt = \int_0^t M_0 g\cos\alpha\sin\alpha\, dt$$

$$\dot{x}\left[\frac{2}{5}(M+M_0) - M_0\cos^2\alpha\right] = M_0 g t\cos\alpha\sin\alpha$$

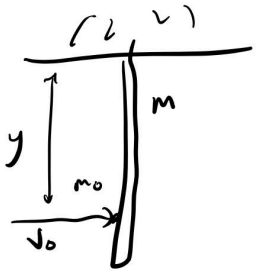
$$\dot{x}(t) = \frac{M_0 g\cos\alpha\sin\alpha}{\frac{2}{5}(M+M_0) - M_0\cos^2\alpha}\, t$$

Question assigned to the following page: 3

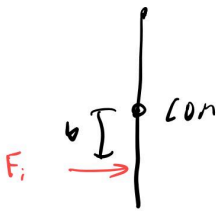
...ivot at one end. Initially it hangs downward and is motionless. Then a particle of mass m_0 which is traveling horizontally with velocity v_0 hits the rod and sticks to it.

a.) Find the distance y from the pivot at which the particle must strike the rod in order that no impulse occurs at the pivot at the instant of impact. *Hint: what is conserved under this condition?*

b.) What is the angular rotation rate of the rod immediately after impact?



$y = b + \frac{l}{2}$



N_0 impulse → NO Force at pivot

$$\Sigma M = I\alpha = \vec{n} = F_i b = I\alpha = I\frac{d\omega}{dt}$$

$$\Sigma F = Ma = F_i = M\frac{dv_{cm}}{dt} \qquad \text{At pivot } v_{pivot} = v_{cm} - \frac{l}{2}\omega = 0$$

$$\frac{dv_{cm}}{dt} - \frac{l}{2}\frac{d\omega}{dt} = \frac{F_i}{M} - \frac{l}{2}\cdot\frac{F_i b}{I} = 0 \therefore b = \frac{2I}{lM} \qquad I_{rod} = \frac{1}{12}Ml^2$$

$$b = \frac{2}{lM}\frac{1}{12}Ml^2 \qquad b = \frac{1}{6}l$$

$$\boxed{y = \frac{2}{3}l}$$

$$I_{pivot} = \frac{1}{3}Ml^2 + \left[m_0\cdot\left(\frac{2}{3}l\right)^2\right] = \frac{1}{3}Ml^2 + \frac{4}{9}m_0 l^2 = \frac{1}{3}\left(M + \frac{4}{3}m_0\right)l^2$$

$$\Sigma M = F_i b = I\frac{d\omega}{dt} \qquad F_i b\,dt = I\,d\omega = \frac{1}{3}\left(M + \frac{4}{3}m_0\right)l^2 d\omega$$

$$\frac{l}{6}\int_0^{\Delta t} F\,dt = \frac{1}{3}\left(M + \frac{4}{3}m_0\right)l^2\int_0^{\omega(\Delta t)} d\omega \rightarrow$$

2 $M_0\Delta V$

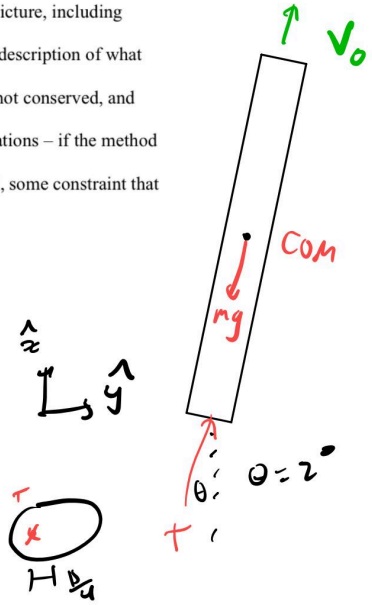
$$2m_0 v_0 = \left(M + \frac{4}{3}m_0\right)l\omega$$

$$6m_0 v_0 = (3M + 4m_0)\,l\omega$$

$$\boxed{\omega = \frac{6m_0 v_0}{(3M + 4m_0)\,l}}$$

Question assigned to the following page: 4

that you chose and began last week. For this week, once again draw the picture, including reference frames and coordinate systems, and provide a concise description of what you are trying to solve. State what quantities are conserved and not conserved, and justify your answers. Attempt a solution using conservation equations – if the method does not work, explain why (i.e. not enough quantities conserved, some constraint that requires another method, etc.).



(not neglecting gravity)

$\vec{r}_0 = [x_0, y_0, z_0]^T$

$\vec{v}_0 = [0, 0, V_0]^T$

with rocket mass M diameter D length L $L >> D$

off axis by $e = \frac{D}{4}$

$x(0) = x_0 \quad y(0) = y_0$

$\dot{x}(0) = 0 \quad \dot{y}(0) = V_0$

$w(0) = 0 \quad \theta(0) = 2°$

what is the maximum altitude the rocket will reach.

Reference/Coord Frames

- Body (Cartesian)
- Inertial (Cartesian)

No this is conserved during burn but every conserved after burn

$L >> D$

$$I = \frac{1}{4}M\left(\frac{D}{2}\right)^2 + \frac{1}{12}ML^2 \simeq \frac{1}{12}ML^2$$

$$\Sigma M = I\alpha = Te \quad \alpha = \frac{Te}{I} = \frac{12\,Te}{ML^2}$$ before burnout

assuming small θ since $\theta_0 = 2°$ and that α is small

$$\Sigma F_z = T\cos\theta - Mg = Ma \quad a = \frac{T}{M}\cos\theta - g$$

$$a = \frac{T}{M}\left(1 - \frac{1}{2}\theta^2\right) - g = \left(\frac{T}{M} - g\right) - \frac{1}{2}\frac{T}{M}\theta^2$$

$$\theta = \theta_0 + \frac{1}{2}\alpha t^2 \rightarrow \theta^2 = \theta_0^2 + \theta_0\alpha t^2 + \frac{1}{4}\alpha^2 t^4$$

$$\therefore \quad a = \left(\frac{T}{M} - g\right) - \frac{1}{2}\frac{T}{M}\left(\theta_0 + \theta_0\alpha t^2 + \frac{1}{4}\alpha^2 t^4\right)$$

$$V_b = V_0 + at_b = V_0\cos\theta_0 + \left\{\left(\frac{T}{M} - g\right) - \frac{1}{2}\frac{T}{M}\left(\theta_0 + \theta_0\alpha t^2 + \frac{1}{4}\alpha^2 t^4\right)\right\} t_b$$

Question assigned to the following page: 4

$$\dots - \frac{T}{2m}\left(\theta_0 t_b + \theta_0 \alpha t_b^3 + \frac{1}{4}\alpha^2 t_b^5\right)$$

↓ integrate

$$z_f = X_0 + V_0 \cos\theta_0 t_b + \frac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\left(\frac{1}{2}\theta_0 t_b^2 + \frac{1}{4}\theta_0 \frac{12Te}{mL^2} t_b^4 + \frac{1}{24}\left(\frac{12Te}{mL^2}\right)^2 t_b^6\right)$$

$$z_b = X_0 + V_0 \cos\theta_0 t_b + \frac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\left[\frac{1}{2}\theta_0 t_b^2 + \frac{3Te\theta_0}{mL^2} t_b^4 + \frac{6T^2e^2}{m^2L^4} t_b^6\right]$$

height at end of burn

$$V_f = V_b - g(t - t_b) = 0 \quad \text{at} \quad t - t_b = \frac{V_0}{g}$$

so $$z_f = z_b + V_b(t - t_b) - \frac{1}{2}g(t - t_b)^2$$

where

$$z_b = X_0 + V_0 \cos\theta_0 t_b + \frac{1}{2}\left(\frac{T}{m} - g\right)t_b^2 - \frac{T}{2m}\left[\frac{1}{2}\theta_0 t_b^2 + \frac{3Te\theta_0}{mL^2} t_b^4 + \frac{6T^2e^2}{m^2L^4} t_b^6\right]$$

Could not use conservation equations because thrust is along the body frame as it rotates, so it is not easy to find energy exerted in x and z axes. z axis is necessary because we want to find max height.

If no initial angle

$$F \cdot t_b = m\,\Delta V \qquad \Delta V = \frac{F \cdot t_b}{m} \qquad T_b = \frac{1}{2}m\left(V_0 + \frac{Ft_b}{m}\right)^2 \qquad T_f = 0$$

$$E_i \neq E_b \text{ but } E_b = E_f \text{ so } \quad \frac{1}{2}m\left(V_0 + \frac{Ft_b}{m}\right)^2 = mg(z - z_0)$$

$$z - z_0 = \frac{1}{2g}\left(V_0 + \frac{Ft_b}{m}\right)^2$$

$$z = z_0 + \frac{1}{2g}\left(V_0 + \frac{Ft_b}{m}\right)^2$$